

T&M equipment can be broadly divided into:

1. Portable instruments like portable V-I signature testers, digital multimeters, portable scopes and other handheld instruments
2. General-purpose bench-top instruments like scopes/counters/power supplies
3. Communication instruments like data communication analysers/radio communication analysers/spectrum analysers, etc
4. Automated test equipment for PCB functional testing, cable harness testing, IC parametric measurements, etc
5. PC-based test instruments (PCI/PXI or LXI-based test and measuring instruments)

the same product is manufactured in India, raw materials such as electronic components and mechanical parts that need to be imported, attract customs duty ranging from 5 to 10 per cent, apart from the CVD. Though the CVD is a modified value-added tax claimable, customs duty is not. Thus local manufacturers lose the cost advantage, despite trimming the foreign exchange outflow and providing local employment. "Also, the interest rates on bank loans are higher now at 13-14 per cent, compared to the 2-5 per cent across southeast Asia," reveals Sabapathi.

What's driving the growth

New advancements in technology give rise to new testing challenges, which in turn create requirements for T&M instruments. These products, once developed, need to be rigorously tested under all scenarios. This testing is usually outsourced to countries like India and Mexico.

The high growth in the communications industry and increased defence spending has stimulated the Indian general purpose (GP) test equipment market. According to Frost & Sullivan, with the escalation of security concerns in India, the government has increased the budget allocation for the defence sector, opening up a huge potential market for GP test equipment.

Also, factors like cost advantages, availability of skilled professionals (with the English language advantage), and India's low cost of living have transformed India into an R&D hub for various industries. These features have resulted in numerous international telecom majors setting up their R&D and

testing facilities in India.

Expansion of communication networks across the country also requires GP test equipment such as spectrum analysers and network analysers. As mobile penetration into rural zones continues to increase, the demand for these testers is also expected to grow.

The Indian wireless test equipment market is also positioned for good growth with investments in R&D and infrastructure development, growth in the mobile phone market and the advent of next-generation wireless technologies. According to a Frost & Sullivan report, Indian base station and air interface test equipment market was worth \$5.4 million in 2007 and is expected to generate revenues of \$9.29 million in 2011. The Indian mobile station test equipment market generated \$18.8 million in 2007. The revenues are expected to grow to \$39.98 million in 2011.

"The penetration and growth of mobile phones is a significant driver for the wireless communication testers market," says Deepa Doraiswamy, programme manager, Frost & Sullivan. "The mobile phone market has registered a phenomenal growth over the last four to five years, resulting in a 275 million subscriber base in 2008. With the addition of close to 10 million us-

**Labour costs,
government policies
and cost of components
are hindering the
growth of T&M
manufacturing in India**

ers each month, the subscriber base is expected to touch 750 million in 2012," she adds.

India is a major hub for electronic manufacturing services (EMS), and the growing demand from this sector will add further impetus to the country's T&M equipment market. Major EMS companies have set up their plants in India to cater to the growing demand of the consumer electronics, telecommunications and automobile sectors.

"With defence and educational sectors becoming increasingly complex, they directly influence the growth of the T&M equipment market in India," says Hitesh Mehta, director, Adtron Technologies.

Application areas

T&M instruments are used for measuring or generating electrical parameters that are used in design, production, troubleshooting or measurement in labs. T&M devices are required during the lifecycle of any product, beginning from the design phase to the development phase and later in the production phase. These equipment cover a host of applications—from the simplest current voltage testing to high-end radio frequency (RF) and microwave measurements—making them integral tools for R&D, manufacturing and installation and maintenance sectors.

Today we have T&M devices that cover a spectrum of equipment implementations, ranging from large automated test equipment (ATE) that evaluate hundreds of devices simultaneously, to small handheld testers that technicians use in the field.

"The major application areas for T&M are the electrical, electronic and telecom industries. The defence sector, R&D labs, railways, aerospace, contract manufacturing, automobile, and education are the main markets for T&M in India," says Sabapathi.

Ambrish Kela, director, Scientech Technologies, adds, "Manufacturing industries, government sectors, defence and academic institutions will continue to be the promising markets